



## Homework 9b: Exercise Set 5.3 Solution

### Question 11

Show that if  $S : \{v_1, v_2, \dots, v_r\}$  is a linearly independent set of vectors, then so is every nonempty subset of  $S$

**Solution:** Solution manual's proof.

11. Suppose that  $S$  has a linearly dependent subset  $T$ . Denote its vectors by  $\mathbf{w}_1, \dots, \mathbf{w}_m$ . Then there exist constants  $k_i$ , not all zero, such that

$$k_1 \mathbf{w}_1 + \dots + k_m \mathbf{w}_m = \mathbf{0}$$

But if we let  $\mathbf{u}_1, \dots, \mathbf{u}_{n-m}$  denote the vectors which are in  $S$  but not in  $T$ , then

$$k_1 \mathbf{w}_1 + \dots + k_m \mathbf{w}_m + 0\mathbf{u}_1 + \dots + 0\mathbf{u}_{n-m} = \mathbf{0}$$

Thus we have a linear combination of the vectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$  which equals  $\mathbf{0}$ . Since not all of the constants are zero, it follows that  $S$  is not a linearly independent set of vectors, contrary to the hypothesis. That is, if  $S$  is a linearly independent set, then so is every non-empty subset  $T$ .

### Question 13

Show that if  $\{v_1, v_2, \dots, v_r\}$  is a linearly dependent set of vectors in a vector space  $V$ , and if  $v_{r+1}, \dots, v_n$  are any vectors in  $V$ , then  $\{v_1, v_2, \dots, v_r, v_{r+1}, \dots, v_n\}$  is also linearly dependent.

**Solution:** From the solution manual.

13. This is similar to Problem 10. Since  $\{v_1, v_2, \dots, v_r\}$  is a linearly dependent set of vectors, there exist constants  $c_1, c_2, \dots, c_r$  not all zero such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_r \mathbf{v}_r = \mathbf{0}$$

But then

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_r \mathbf{v}_r + 0\mathbf{v}_{r+1} + \dots + 0\mathbf{v}_n = \mathbf{0}$$

The above equation implies that the vectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$  are linearly dependent.

### Question 15

Show that if  $\{v_1, v_2\}$  is linearly independent and  $v_3$  does not lie in  $\text{span}\{v_1, v_2\}$ , then  $\{v_1, v_2, v_3\}$  is linearly independent.

**Solution:** From the solution manual.

15. Suppose that  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$  is linearly dependent. Then there exist constants  $a, b$ , and  $c$  not all zero such that

$$(*) \quad a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 = \mathbf{0}$$

Case 1:  $c = 0$ . Then  $(*)$  becomes

$$a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{0}$$

where not both  $a$  and  $b$  are zero. But then  $\{\mathbf{v}_1, \mathbf{v}_2\}$  is linearly dependent, contrary to hypothesis.

Case 2:  $c \neq 0$ . Then solving  $(*)$  for  $\mathbf{v}_3$  yields

$$\mathbf{v}_3 = -\frac{a}{c}\mathbf{v}_1 - \frac{b}{c}\mathbf{v}_2$$

This equation implies that  $\mathbf{v}_3$  is in  $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ , contrary to hypothesis. Thus,  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$  is linearly independent.

### Question 18

Under what conditions is a set with one vector linearly independent?

**Solution:** The vector  $\vec{v}_1$  should be nonzero. Let's take zero vector...

$$\vec{0} = k_1 \vec{v}_1 \Rightarrow k_1 = 0 \quad \because \vec{v}_1 \neq 0$$

### Question 24

Indicate whether each statement is always true or sometimes false. Justify your answer by giving a logical argument or a counterexample.

(a) *The set  $2 \times 2$  of matrices that contain exactly two 1's and two 0's is a linearly independent set in  $M_{22}$ .*

**Solution:** False. Let the set of  $2 \times 2$  matrices be

$$\left\{ E_1 = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, E_4 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$$

$$\vec{0} = k_1 E_1 + k_2 E_2 + k_3 E_3 + k_4 E_4$$

Take  $k_1 = k_3 = 1, k_2 = k_4 = -1$  Hence it is not linearly independent.

(b) *If  $\{v_1, v_2\}$  is a linearly dependent set, then each vector is a scalar multiple of the other.*

**Solution:** False. If  $v_1 = 0$  but  $v_2$  is not the zero vector, then  $v_1 = 0v_2$  (so  $v_1, v_2$  is dependent) but  $v_2$  is not a scalar multiple of  $v_1$ .

(c) *If  $\{v_1, v_2, v_3\}$  is a linearly independent set, then so is the set  $\{kv_1, kv_2, kv_3\}$  for every nonzero scalar  $k$ .*

**Solution:** True. By definition of linear independent, any vector  $v_i$  can be written as the linear combination of the  $\{kv_1, kv_2, kv_3\}$ .

(d) *The converse of Theorem 5.3.2a is also true.*

**Solution:** False. Let's we have linearly dependent set  $V = v_1, v_2$  and  $v_1$  is any scalar multiple of  $v_2$ .

### Question 25

**Show that if  $\{v_1, v_2, v_3\}$  is a linearly dependent set with nonzero vectors, then each vector in the set is expressible as a linear combination of the other two.**

**Solution:** False. The definition only requires that at least one vector be expressible as a linear combination of the other two. Here's an easy example in  $R^2$  that shows we don't have to have each vector expressible in terms of the other two (even when all the vectors are nonzero):  $v_1 = (1, 0), v_2 = (2, 0), v_3 = (0, 1)$ . Obviously the vector  $v_3$  cannot be expressed in terms of  $v_1$  and  $v_2$ .